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A review of hazards associated with primary lithium and lithium-ion batteries

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ABSTRACT

Primary lithium batteries contain hazardous materials such as lithium metal and flammable solvents, which can lead to exothermic activity and runaway reactions above a defined temperature. Lithium-ion batteries operating outside the safe envelope can also lead to formation of lithium metal and thermal runaway. Despite protection by battery safety mechanisms, fires originating from primary lithium and lithium-ion batteries are a relatively frequent occurrence.

This paper reviews the hazards associated with primary lithium and lithium-ion cells, with an emphasis on the role played by chemistry at individual cell level. Safety mechanisms to prevent the occurrence and limit the consequences of incidents are reviewed, together with safety tests to monitor compliance with battery safety regulations and standards. Incident information from news accounts and open literature sources are reviewed to extract causal information.

It is concluded that the potential severity of incidents during storage, transport and recycling of waste batteries can be significantly higher than in end-use applications. Safe storage, packaging and labelling practices, as well as communication among the parties involved, are essential to ensure safety across the battery lifecycle. It is recommended that a database of lithium battery incidents would be valuable to improve the evidence base for informing accident prevention measures.

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1. Introduction

Since the 1990s, primary lithium and lithium-ion batteries have become the power supply of choice in many consumer, industrial and military applications due to their advantages in terms of energy density over other battery technologies. A considerable research effort has been invested in developing inherently safer chemistries, and the development of more effective thermal management systems and protection devices (Balakrishnan et al., 2006; Hassoun et al., 2009).

The aim of this paper is to review the safety characteristics of commercial primary lithium and lithium-ion battery technologies, focusing on side reactions and thermal behaviours that could compromise safety during operation. Safety mechanisms and tests that have been designed to evaluate battery performance during normal use and abuse circumstances are

subsequently reviewed in light of experimental evidence available in the open literature.

Hazards associated with primary lithium and lithium-ion cells have materialised not only during use at the intended application, but also during transport and storage of new and used battery packs; or when end-of-life batteries undergo treatment for recycling to recover marketable materials or to meet the requirements brought by legislation. A number of recent incidents during transport, storage and recycling operations described in the news accounts have also been reviewed. Although definitive evidence on the actual mechanism initiating the events is often lacking, incidents can at times be linked to incorrect handling, storage and packaging practices that may lead to mechanical damage, water ingress, and/or internal or external short-circuit of charged batteries.

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2. Hazards associated with primary lithium cells

Primary lithium batteries are high-energy systems that find applications in military and industrial systems, although they have also been used in consumer applications. Primary (non-rechargeable) lithium batteries contain metal lithium as anode material, flammable or highly flammable organic solvents, and potentially explosive components (perchlorates such as a lithium perchlorate electrolyte), although the use of the latter is in decline. Examples of materials used in the positive electrode are manganese dioxide, poly(carbonmonofluoride), iron disulphide, vanadium pentoxide, copper oxide, copper oxyphosphate or thionyl chloride (Levy and Bro, 1994). Safety issues may arise during the life cycle of primary lithium batteries due to any of the following processes:

- Highly flammable hydrogen gas is generated, usually followed by ignition, upon contact of lithium metal with water. Water may originate from inadequately dried cell components or ingress upon loss of mechanical integrity of the cell (Lim et al., 2007).
- Formation of dendrites and eventual internal short-circuit of the cell. Dendrite formation has long been an issue if batteries containing lithium metal are operated as rechargeable systems (Park et al., 2008).
- Thermal effects/exothermic activity. This can lead to thermal runaway if unmanaged by internal or external safety devices. High-energy primary lithium batteries based on thionyl chloride have been reported to experience self heating at temperatures as low as 148 °C with adiabatic temperature rises of 14 °C and self-heating rate of approximately 5 °C/min (Mores and Ottaway, 1998). As the temperature increased above 172 °C, heating rates of nearly 11 °C/min have been reported.
- Oxidation of organic solvents by lithium perchlorate. Perchlorates are strong oxidising agents known to give rise to explosive reactions with organic substances (Levy and Bro, 1994).

3. Hazards associated with lithium-ion cells

Lithium-ion cells have long been considered as an inherently safer alternative to lithium metal-anode cells because lithium metal is not re-deposited upon charging under normal circumstances (Balakrishnan et al., 2006). Numerous materials for positive and negative electrodes have been proposed over the years; a large proportion of conventional lithium-ion cells use a carbon-based anode with the positive electrode being a metal oxide that contains lithium such as LiCoO_2 . The electrolyte consists of a lithium salt (e.g. LiPF_6) that is able to dissolve in organic solvents such as methylcarbonate or diethylcarbonate. Novel materials for electrolyte solutions, positive and negative electrodes (Fergus, 2010; Gulbinska et al., 2011) are constantly being developed to improve not only electrochemical performance parameters such as capacity, energy density and cycle life, but also safety performance. Hazards associated with lithium-ion cells can originate from the following side reactions:

- Molten lithium can form in the event of overcharging metal lithium cells due to the low melting point of lithium metal

(180 °C). However, this is less likely to occur in lithium-ion cells as metal lithium is replaced by lithiated carbon materials.

- Reactions between the organic solutions and the electrode surface occur when the temperature of the cell increases, particularly if the solid-electrolyte interface (SEI) is disrupted. The SEI is the contact surface between the electrolyte solution and the Li_xCoO_2 electrode and consists of insoluble products generated as the battery is charged for the first time (Doh et al., 2009). This interface becomes unstable when the cell temperature rises above 70–100 °C and decomposes exothermically.
- Heat generation and thermal management are critically important for the safe operation of lithium-ion cells. Both reversible and irreversible heat generation must be considered in the battery management system as reviewed by Viswanathan et al., 2010. Internal resistance is responsible for irreversible heat, whilst reversible heat is linked to the reduction reaction that takes place in the positive electrode and the heat generated at the negative electrode. Significant differences in reversible heat effects across the various lithium-ion chemistries: LiCoO_2 cells produced reversible heat of 700% the irreversible values with heat generation rates difference between charge and discharge of 1.4 kW (5–40% charge). The value is significantly lower, approx 50 W, for LiFePO_4 between 2 and 95% state of charge.

3.1. Thermal runaway of lithium-ion batteries

Thermal runaway of lithium-ion batteries has been described as a three-stage process (Abraham et al., 2006; Selman et al., 2001):

- (1) Anodic reactions start at about 90 °C. This is the rate-limiting step. As temperature rises above 120 °C, decomposition of the solid electrolyte interface (SEI) layer follows, leading to reduction of the electrolyte at the lithiated graphite negative electrode.
- (2) In a second step of the thermal runaway mechanism, exothermic reactions at the positive electrode start as the temperature rises over 140 °C. Oxygen rapidly evolves at this stage.
- (3) The positive electrode decomposes and the electrolyte gets oxidised at temperatures above 180 °C. This is a high-rate exothermic process with a temperature rise as high as 100 °C per min.

Thermal runaway and heat effects in lithium-ion cells are sensitive to the state of charge (the higher the charged voltage the lower the onset temperature) and also depend, for the same cell type, on the history of the cell and the load applied.

Tests performed by Zhang et al. (1998) discovered that lithiation of the positive electrode resulted in better thermal stability. The composition of the electrolyte also had an impact on the reported thermal stability (ethylene carbonate/dimethyl carbonate being more stable than EC/DME). The positive electrode material also affected thermal stability: cells containing LiCoO_2 or LiNiO_2 performed worse than LiMn_2O_4 . Abraham et al. (2006) reported thermal stability studies using accelerating rate calorimetry on $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ electrodes: self-heating reactions started as the cell reached 84 °C and progressed until the cell was quenched at 150 °C.

Alternative materials for the positive electrode are used to prevent the problems associated to LiCoO_2 :

- Coating the positive electrode material with inert oxides can limit direct contact of LiCoO_2 with the electrolyte and improve thermal stability. Other examples are $\text{Li}_4\text{Ti}_5\text{O}_{12}$ (Yi et al., 2010), or LiFePO_4 (the latter for $\text{Li}[\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}]\text{O}_2$ -based electrodes as suggested by Kim et al. (2010)).
- Improved thermal stability linked to positive electrodes different from LiCoO_2 such as the $\text{Li}_4\text{Ti}_5\text{O}_{12}/\text{LiMn}_2\text{O}_4$ electrode reported by Belharouak et al. (2007). Cells showed no thermal runaway up to temperatures of 400°C . No fires or explosions occurred although the cell vented at 150°C , due to pressurisation of the organic solvents. The effect of the degree of lithiation on stability lithium-ion cells based on Li_xNiO_2 , Li_xCoO_2 and $\text{Li}_x\text{Mn}_2\text{O}_4$ was studied by Zhang et al. (1998). Onset temperatures decreased as x decreased. $\text{Li}_x\text{Mn}_2\text{O}_4$ showed an exotherm at 225°C but energy released was much lower and less dependent on the degree of lithiation (x).

3.2. Electrolyte decomposition and evolution of gas species from lithium-ion batteries

Thermal effects in lithium-ion cells can lead to gas generation and cell overpressurisation, and eventually, can trigger venting or rupture of the battery. The composition of gas species from overcharged lithium-ion batteries is available from Ohsaki et al. (2005). CO_2 and CO were found to evolve from the positive electrode, with a small fraction of methane ($<10\%$) also being generated. This confirmed that CO_2 is generated by oxidation of the electrolyte at the positive electrode rather than by reactions in the carbonaceous negative electrode. Highly flammable gas products can also evolve: H_2 was the main component in the gas evolved at the negative electrode (ca. 50%) with small amounts of CH_4 , C_2H_4 , C_2H_6 .

LiPF_6 in the electrolyte decomposes to form LiF and PF_5 , with PF_5 acting as a catalyst for solvent decomposition (Abraham et al., 2006). Products of electrolyte decomposition have been listed as CO_2 , C_2H_4 , fluoroethane ($\text{C}_2\text{H}_5\text{F}$), diethyl ether ($(\text{C}_2\text{H}_5)_2\text{O}$), highly toxic alkylfluorophosphates ($\text{OPF}_2(\text{OCH}_2\text{CH}_3)$, $\text{OPF}_2(\text{OCH}_2\text{CH}_3)_2$) and fluorophosphoric acids (OPF_2OH , $\text{OPF}(\text{OH})_2$) for 1M LiPF_6 in diethyl carbonate or mixtures with dimethyl carbonate (Campion et al., 2004). Products of the electrolyte and solvent decomposition reactions may react with the electrode surface to form CO_2 , oxalates carbonates (at the negative electrode) or complex fluorinated oligoethylene oxides $\text{OPF}_2\text{OCH}_2\text{CH}_2\text{F}$.

Evidence of the electrolyte decomposition mechanisms is also available from Onuki et al. (2008). $\text{LiNi}_x\text{Co}_y\text{Al}_{1-x-y}\text{O}_2$ (LNCAO) electrodes in Li-ion cells generate larger amounts of CO_2 than the conventional LiCoO_2 electrode. Ethylene carbonate (EC) was the main source of CO_2 , whilst CO appears to be formed from solvent and non-solvent components. Formation of C_2H_6 was linked to EC, whereas C_2H_4 was produced by diethyl carbonate (DEC). Kong et al. (2005) reported similar gas generation patterns from lithium-ion batteries having LiCoO_2 , LiMn_2O_4 and LiFePO_4 electrodes. However, $\text{LiMn}_2\text{O}_4/\text{C}$ and LiFePO_4 systems were found to generate larger amounts of C_2H_2 than LiCoO_2 as the voltage increased.

3.3. Safety tests

All battery designs are subjected to safety tests to evaluate performance in the event of faults developing. Experimental evidence available in the open literature suggests markedly different behaviours under test conditions among lithium-ion cells depending on cell chemistry. A list of safety tests routinely performed on lithium-ion batteries is available from Spotnitz and Franklin (2003).

Oven test: Consumer batteries are heated to 150°C in an environmental chamber (e.g. according to the procedure described in UL 20541(1997)).

Overcharge test: The cell is charged to a chosen voltage above the recommended value. Results of overcharge tests on prismatic lithium-ion cells are available from Leising et al. (2003); low rate charging of the cells did not result in thermal runaway but at higher rates (9C; C represents a charge or discharge rate equal to the capacity of a battery divided by 1 h), thermal runaway occurred particularly when the positive electrode was de-lithiated. At temperatures above 190°C rupture of the cell occurred, due to melting of the metallic lithium formed at the negative electrode. Tests in Tobishima et al. (1997) showed similar behaviours: cells did not vent when overcharged at lower rates, but ignited at higher rates (2C). Ohsaki et al. (2005) proposed an overcharge reaction mechanism for prismatic LiCoO_2 cells according to a four-step process:

- (1) Migration of lithium from the negative to the positive electrode occurs as the cell voltage increases. Gas evolution and temperature rise in the outer casing is low.
- (2) As overcharge approaches 100%, cell impedance and positive electrode resistance increase, resulting in heat generation due to the Joule effect and the exothermic decomposition of the electrolyte mixture. The electrolyte is decomposed by Co^{4+} ions whilst the remaining LiCoO_2 rhombohedral structure evolves to an electrochemically inactive monoclinic structure.
- (3) Rapid temperature rise due to reaction between the lithium-deprived positive electrode and the electrolyte. The reaction accelerates as the cell's internal temperature rises above 60°C and CO_2 gas is generated.
- (4) As the temperature approaches $130\text{--}135^\circ\text{C}$, the cell may behave according to one of the following pathways:
 - (a) If the polyethylene separator shutdown occurs, which is most likely in the event of a low rate ($<1\text{C}$) overcharge, the temperature would decrease with no rupturing.
 - (b) Higher rates of overcharge could lead to thermal runaway and eventual cell rupture/venting as a highly exothermic reaction between the negative electrode (with deposited lithium) and the electrolyte occurs.

In experiments performed by Saito et al., the heat generated when the cell was overcharged was proportional to cell charge. The negative electrode was found to be responsible for thermal runaway in overcharged cells, in agreement with the observations made by Leising et al. (2003). The onset temperature for thermal runaway was 180°C in undercharged conditions but only 80°C when the cell was overcharged.

Short-circuit: Short-circuit tests are usually performed by connecting a low resistance element to the terminals of the battery. The heat generation mechanism is by current flow through the battery. The overall heating experienced by the

cell will primarily depend on the current flow as it is discharged, but also on the ability of the external circuit attached to the battery to dissipate heat.

Nail test: In this test, the cell is punctured by a nail driven from the surface of the cell at a constant speed (normally 8 cm s^{-1}). There are two components to the heat generation as the cell is discharged: firstly, from current flowing through the cell as the separator integrity is broken and, secondly, from the current that flows through the nail. Heating of the cell is localised in and around the nail as opposed to the more uniform distribution that usually takes place in a conventional short-circuit test. Thermal runaway during a nail test is more likely to occur when the cell is punctured by the nail to relatively shallow depths (Spotnitz and Franklin, 2003; Wu et al., 2004). As the contact area is relatively small in shallow depth punctures, heat dissipation will be limited. Exothermic reactions of electrolyte and electrode decomposition/oxidation are more likely to occur as they are initiated at relatively high temperatures. In the case of deep punctures of cells, the contact area is higher and therefore heat dissipates more easily.

Spotnitz et al. (2003) also reported nail test results performed by Dahn on 18,650 LiCoO₂ cells:

- When puncture of the cell is performed at a low rate, temperatures in the nail may reach values as high as 600 °C. Fast, deep punctures resulted in nail temperature below 140 °C.
- Nail tests and crush tests on overcharged cells resulted in smoking/fires, as opposed to no overheating observed when 720–835 mA h prismatic cells underwent equivalent tests under normal state of charge.

Crush test: The battery is compressed by applying a mechanical force. As the test progresses, the cell undergoes discharge, which is a heat-generating process in itself, and, as the mechanical integrity of the cell is broken, short-circuit between the electrode terminals occurs. Although a somewhat uniform discharge of the battery is expected, heat can be generated and dissipated locally in the area where the short-circuiting is taking place.

The outcomes of safety/abuse tests on some lithium-ion batteries performed following the IEC 62133 standard have been made publicly available. Eom et al. (2007) reported the results of these tests on an unspecified lithium-ion battery (3.7 V nominal voltage and 1000 mA h). The cells were charged and discharged to determine failure time and number of charge and discharges before failure. Cells performed well in the overcharge test (no smoke, fire or explosion occurred when they were charged up to 250% capacity). The cell temperature reached 90 °C during external short circuit, which was considered acceptable. Internal short circuit was tested by nail penetration (5 mm diameter) at 1 cm s^{-1} through the electrode of a fully charged cell. Cell temperature increased to 82 °C. Thermal exposure was tested at 130 °C for 30 min; gas generation occurred as batteries swelled, although no venting, fire or explosion occurred. All these outcomes can be seen as satisfactory responses to abuse tests.

Kitoh and Nemoto (1999) performed short-circuit, overcharge, nail penetration and oven tests on 100 Wh LiMnO₄/carbon batteries. Oven tests at 130 °C showed exothermic activity with self-heating of the cell to over 190 °C after 80 min. There was venting but not ignition. Melting of the separator occurred after 40 min of the test.

Govar and Banner (2003) reported tests performed on three types of military lithium-ion batteries tested to overcharge

conditions. Although all batteries were deemed suitable for the intended applications, outcomes of tests significantly differed among the three types. Venting and/or fire were observed in some cases although the vented material remained within the battery casing. In other cases, no venting was observed during overcharge, overdischarge or short-circuit tests even with safety mechanisms being bypassed. Thermal abuse tests resulted in venting at temperatures between 170 °C and 290 °C accompanied by missile/flying debris generation.

A list of possible exothermic reactions during abuse test conditions on lithium-ion batteries is available from Spotnitz and Franklin (2003):

- (1) Decomposition of the solid electrolyte interface that protects the negative electrode from reaction with the solvent. This layer can decompose generating heat at temperatures above 90 °C.
- (2) Reaction of the lithium intercalated in the negative electrode with the electrolyte mixture. This can occur at temperatures above 120 °C (the solid electrolyte is most likely to have failed at this temperature).
- (3) Fluorinated binders used in the electrodes, such as PVDF, can react exothermically with lithiated carbon in the negative electrode.
- (4) Exothermic decomposition of the electrolyte can occur at temperatures above 200 °C.
- (5) Exothermic decomposition of the positive electrode in an oxidised state can produce oxygen that reacts with the electrolyte.
- (6) Reaction of metallic lithium deposited in the negative electrode with the electrolyte (overcharge conditions).
- (7) Heat is released upon cell discharge.

New and old batteries have been reported to show significant differences in safety performance under tests conditions (Wu et al., 2004). Abuse tests on used batteries may be therefore more suitable to characterise hazards associated with used and waste batteries.

3.4. Safety mechanisms in lithium-ion batteries

Safety mechanisms are in place to prevent the occurrence and limit the consequences of malfunctions in battery systems. According to Rosenkranz (2010), safety in complex battery systems is provided at three different levels as shown in Table 1:

- At battery hardware level, safety mechanisms involve cell design features such as safety vents, shutdown additives, current cut-off device and separator materials (Balakrishnan et al., 2006). As highlighted by Takami et al. (2009), these are essential to prevent the occurrence and/or limit the extent of consequences of internal malfunction at individual cell and battery pack level. The system must be designed in a way that any material vented is safely contained and/or directed to the surroundings.
- At system hardware level, adequate electronic control to prevent overcharge, over discharge and overheating of the battery packs is necessary, including electronic balancing to prevent unbalanced states of charge among packs. The electrical hardware is essential to provide safety at system level; fuses are needed to protect against high current excursions in the system performance, and adequate contactors minimise the possibility of external short-circuit. There is also a

Table 1 – Overview of the safety concept for battery modules in automotive applications (from Rosenkranz, 2010).

System software	System hardware	Cell hardware
Measurement of battery system characteristics	Electronics hardware	Cell-level design features
Cell/pack voltage	Over and under voltage protection	Pressure vent
Temperature	Over temperature	Current interrupt device
Device feedback	Cell balancing circuitry	Separator materials
Sensor validity	Electrical hardware	System structure
Consistency check	Fusing for over-current protection	Contain any vented materials
Fault or Failure	Contactors	
Control actions	Mechanical hardware	
	Optimum thermal management	
	Structural protection	

mechanical aspect to provide safety at the system hardware level; batteries must be provided with structural protection as well as a thermal management system (e.g. adequate ventilation) to prevent overheating due to operation or heat input from the surroundings. Large capacity battery modules are constructed by densely packing small commercial cells. Due to the high specific energy and thermal behaviour of 18,650 Li-ion cells modules should take into consideration heat dissipation issues that arise in high-voltage and/or high-capacity module applications. The safety of densely packed battery packs and modules may be compromised by a defect or thermal runaway affecting an individual cell within the pack. Thermal management of the group of cells forming the pack and module is necessary to prevent propagation of these thermal effects. Battery packs and modules are frequently thermally managed using air-cooling systems. However novel thermal management systems based on phase change materials have been proposed (Kizilel et al., 2009).

- Last, but not least, safe operation of battery systems requires controls at the software system level. Measurement of the battery performance is necessary to ensure safe operation. Examples of parameters that may be good indicators of cell performance are cell/pack voltage, temperature, current, state of charge, and their values may differ among cells in a pack or module. Cell capacity and state of charge of each cell within a module can differ, despite being constructed with cells that should be theoretically identical and have seen an equivalent charge history. Controls at the system software level therefore allow consistency checks and detection of faults or failures.

3.4.1. Positive temperature coefficient device (PTC)

PTC devices work by responding to increased currents passing through an element. Heating up of the device occurs due to the Joule effect, which is translated into an increase on the resistivity of the device, which in turns would limit the current flow.

Kise et al. (2007) described the use of internal PTC components for the positive electrode material of the cell based on carbon black/polyethylene mixtures coated on the positive electrode together with acetylene black. Internal PTC elements are considered safer than external PCT devices as external PTC devices cannot prevent the occurrence of internal short-circuit. Cells with a conventional LiCoO₂ positive electrode containing 12% of the PTC were overcharged up to 10 V did not result in fire or explosion.

PTC devices in 18,650 cells consist of two metal-coated sides that confine the polyethylene polymer mixed with carbon black. The PTC devices can be characterised via a number

of tests, such as those performed by Smith et al. (2010). These authors determined the resistance of the cell headers by applying current pulses as the temperature of the PTC was increased when placed in an oven/environmental chamber. As the cell was shorted, the PTC device eventually tripped and heated up whilst carrying the majority of the current.

3.4.2. Redox shuttles for lithium-ion batteries

According to Chen et al. (2009), a redox shuttle is an electrolyte additive that can be reversibly oxidised/reduced at characteristic potential and provides an intrinsic overcharge protection for lithium-ion batteries that neither increases the complexity and weight of control circuit nor permanently disables the cell when activated. This last statement marks the difference between redox shuttles and inactivation agents/thermal runaway inhibitors. Inactivation agents are based on polymerisation of the additive or electrochemical reactions forming copious amounts of gas that triggers cell venting leading to irreversible inactivation of the individual battery cell and the whole battery pack.

According to Chen et al. (2009) redox shuttles for good overcharge protection should have the following characteristics:

- Stable after hours of operation;
- Redox potential between 0.3 and 0.4 V above the operating potential of the positive electrode.

As reviewed by Chen et al. (2009), redox shuttle additives for overcharge protection of 3 V lithium-ion batteries are mainly I[−], Br[−] ions and ferrocene/ferrocene-based compounds such as N-butylferrocene, N,N-dimethylferrocene, 1-acetylferrocene, or ferrocenemethanol. The main characteristic of these substances is that their redox potential versus the Li⁺/Li pair is slightly above 3 V and therefore they will carry a considerable amount of charge between the positive and negative as the cell voltage rises above 3 V.

Other redox shuttles are available from Chen et al. (2005): aromatic compounds suitable for LiFePO₄ electrodes in cells operating at voltages below 4 V are 2,5-di-terbutyl-1,4-dimethoxybenzene and 10-methylphenothiazine. Redox shuttles suitable for operating with LiCoO₂ and LiMn₂O₄ (at voltages over 4.2 V) are, for instance, 2-(pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole and non-aromatic Li₂B₁₂F₁₂ with a redox potential of 4.74 V, and capable of providing overcharge for all positive electrode materials.

Li-polymer batteries based on LiCoO₂ electrodes with methacrylate and azo compounds in the gel electrolyte have been demonstrated to have an overcharge protection mechanism similar to the gas recombination shuttle mechanism for sealed lead-acid, Ni–Cd and Ni–MH batteries (Nozu et al.,

2006). O₂ and CO₂ generated at the overcharged positive electrode (Li_{0.62}CoO₂) migrated through the electrolyte and reacted with the lithium in the surface of the C₆Li negative electrode to form lithium oxide and carbonate Li₂O, Li₂CO₃. The authors acknowledged that the kinetics of the process are not fast but suggested this mechanism as a possibility to reproduce, in lithium-ion batteries, the gas recombination effect that takes place in sealed Ni–Cd and Ni–MH batteries.

3.4.3. Thermal runaway inhibitors/inactivation agents

Fire retardant compounds are added to lithium-ion battery electrolyte solutions to suppress thermal runaway reactions if the temperatures or conditions that give rise to these events are reached at any point during service. A number of materials have been proposed: Mandal et al. (2006) studied three aromatic esters that contained phosphorous, compounds that are known to suppress thermal runaway in lithium-ion batteries. The phosphate group of the molecule acted as flame-retardant in itself, whilst the aromatic functionality generated free radicals that limit oxygen availability and suppressed flame propagation if ignition was to occur.

4. Incidents involving primary lithium and lithium-ion batteries during transport, storage and treatment operations

Despite the systems in place to ensure safe operation of lithium and lithium-ion cells, incidents involving primary lithium and lithium-ion batteries have occurred relatively frequently and have attracted a great deal of media coverage in recent years. Incidents during battery use were, in many cases, followed by widespread recall of defective battery packs (Balakrishnan et al., 2006). Design errors, manufacturing faults or operation outside the safe envelope can not only cause shortened battery life but also, more significantly, thermal management problems that may lead to overheating and, if the safety mechanisms in the battery fail, to fire and explosion events. According to the US Consumer Product Safety Commission (Consumer Product Safety Commission (CPSC), 2009), as highlighted by Reif et al. (2010), tens of millions of primary lithium and lithium ion batteries have been recalled since introduction of these technologies in the early 1990s.

Incidents involving primary lithium and lithium-ion batteries in transport, storage, during sorting, mechanical treatment or undergoing recycling process for recovery of marketable materials are a relatively less frequent occurrence although examples can be found in the news accounts.

A fire in Clarence, New York (US) on August 14, 2002 burned a total of 68 tonnes of waste, two thirds of them being lithium batteries (Robbins, 2008). It was reported that the fire began in the lithium treatment area as employees were opening a 55-gallon drum of used lithium batteries for recycling by thermal treatment. It was believed that shorts between batteries contained in the drum initiated the event. A similar fire at a lithium battery recycling plant took place in Thorold (Canada) in February 2007 (Downs, 2007). The premises stored waste lithium batteries in closed containers inside a dry warehouse. It was reported that the fire started at one of the waste battery containers, possibly due to water ingress either via condensation or snow. No information on the type of container was available from the news account. It is not known if the containers were suitably weatherproof for storing lithium batteries. It was suggested that condensation could have been caused

by temperature changes during battery transport and storage (battery containers were moved into the heated warehouse from a cold loading dock).

The use of unsuitable containers to store waste primary lithium batteries has also been suggested as the most likely cause of the fire that broke out at a waste treatment plant near Preston (UK) in July 2007 (HSE, 2009). The fire started in an open area of the site where primary lithium batteries were being stored in yellow clinical waste containers that were not fully watertight. It appears that the batteries had not been packed properly, as there was no inert packing material and the battery terminals were not taped. These measures are necessary to minimise the risk of spontaneous ignition of mechanically damaged cells or short-circuits. The waste batteries had been placed near drums containing waste flammable materials. The company was fined £150,000 for breaches of health and safety regulations after successful prosecution instigated by HSE.

Recurrence of lithium battery fires is illustrated by some events, such as a series of fires at a large battery recycling plant in Trail (British Columbia). The site processes primary lithium batteries as well as other chemistries. Back in 1995, 40 tonnes of batteries burned at the plant according to CBC news (2009). This event led to the implementation of a material segregation approach to minimise the hazards of primary lithium battery recycling on the site: primary lithium cells were placed in smaller earth mounded bunker-type structures to prevent the spread of fire to the whole facility. Despite this, a total of three fires occurred in 2000, with at least one of them started by primary lithium battery recycling operations. A fire at the premises was also reported in 2008. In the last occurrence, in 2009, primary lithium batteries stored in one of the earth-mounded bunker structures caught fire. It was reported that the most likely cause of the fire was an internal short in one of the batteries stored. Flaming lasted for a total of 22 h with fire fighting having to stay on hold during the fire due to projectiles being ejected from the battery store. As the fire developed, local residents were evacuated with other nearby residents were told to stay indoors due to the smoke plume containing sulphur dioxide. Despite primary lithium battery segregation, the fire spread to a nearby local recycling plant by the flaming projectiles ejected from the battery plant (CBCnews, 2009).

Incidents involving primary lithium and lithium-ion batteries have been documented in greater detail when high reliability industries such as aircraft transport were involved. Incidents during aircraft transport of primary lithium and lithium-ion batteries were reviewed in the National Transportation Safety Board (NTSB) incident report (NTSB, 2007). According to this US FAA document, 82 incidents involving all types of batteries were reported within the aviation industry. Fourteen of the incidents involved lithium-containing batteries and 13 of these incidents involved primary lithium batteries. Two incidents had serious consequences and resulted in stricter safety regulations for the aircraft transport of lithium batteries (ICAO, 2010; IATA, 2010); these were the incident at Los Angeles International Airport (LAX) in 1999 and the incident at Philadelphia International Airport in 2006.

The incident at Los Angeles International Airport involved two pallets of primary lithium metal batteries that were being transported by Northwest Airways personnel using fork-lift trucks (Farrington, 2001). One pallet contained a total of 100,000 primary lithium cells (Li/MnO₂-type) whilst the other had 20,000 of primary lithium cells together with other

rechargeable cells. The ignition mechanism has not been established with a high level of certainty, but the following causes have been suggested: mechanically damaged battery packages, contact between loose battery packs that may have led to forced charge, discharges, or short-circuiting. The cargo was not considered as dangerous goods at the time due to the small amount of lithium metal. However, the fire could not be put out by conventional fire fighting equipment available to staff.

Aircraft transport of lithium-ion batteries has also been subjected to stricter conditions after a fire on a cargo aircraft approaching Philadelphia International Airport (2006), which was transporting lithium-ion batteries. Although the cause of the fire was not fully established as a result of the investigations, the US Department of Transport and the FAA issued stricter standards for the transport of lithium-ion batteries (FAA 2006, DOT, 2007 and 2009 in Reif et al., 2010). More recently, a cargo flight transporting lithium-ion batteries crashed in Dubai after a fire broke out (Shaheen, 2010). This has, again, attracted a great deal of media attention and calls for closer monitoring of primary lithium and lithium-ion battery transportation.

Following the aforementioned incidents during transportation of lithium batteries, technical instructions for the Safe Transport of Dangerous Goods by Air were issued by ICAO, the International Civil Aviation Organisation, which is the United Nations agency for the codification of the principles for air transportation. A guidance document covering the transport of lithium batteries was issued (ICAO, 2010), which covers a number of revisions to earlier regulations, in particular, packing instructions according to the cell chemistry in order to prevent external short circuit.

Primary lithium batteries are not allowed for transport on passenger aircraft unless they meet Packaging Instructions 969 or 970, which states no extra cells apart from the ones needed for powering the device, the lithium content when at 100% SOC is not more than 5 g, the aggregate lithium content of the negative electrode is not more than 25 g when fully charged, and the total net weight of lithium batteries is not above 5 kg.

Packaging instructions 969 and 970 refer to the cells needing to meet UN Manual of Test and Criteria, part III, section 38.3, to have a safety vent that prevents explosion and means to prevent external short circuits. Battery packs with cells connected in parallel must contain means to prevent reverse current flow (diodes, fuses). Transport of cells containing positive electrodes with SO₂, sulphuryl chloride or thionyl chloride discharged to open circuit voltages below 2 V or under two thirds of the voltage of the cell at 100% SOC are forbidden from transport.

The United Nations' recommendations require that all primary lithium and lithium-ion batteries produced from 2003 onwards pass a series of tests prior to transport:

- Altitude simulation
- Thermal test

- Vibration
- Shock
- External short circuit
- Impact
- Overcharge
- Forced discharge

In the UK, the Department for Transport has issued Authorisation 214 for the carriage of up to 333 kg of lithium-ion batteries (The Department for Transport, 2009). In brief, lithium-ion batteries must be mixed with other portable batteries (i.e. not segregated) and kept tightly packed in a one-use plastic liner placed within a plastic or metal container. It is not known how this requirement could be complied with in a scenario of large-scale implementation of lithium-ion battery-powered vehicles. The large inventories of new and waste lithium-ion batteries can make mixed transport impracticable. It follows that further regulations will be needed in order to ensure the safety and feasibility of transport of waste batteries.

Establishing the ultimate cause of a fire involving lithium batteries is inherently difficult as forensic evidence at the seat of the fire is usually severely damaged by heat and/or ejected as projectiles after over pressurisation of battery packs. Despite this, there would still be value in recording any evidence that may be available from the investigation and this information disseminated. However, there is no common source of information to document incidents involving lithium batteries. Such a database would provide an invaluable insight into earlier incident occurrences, facilitating statistical and causal analysis of the events so that lessons are learnt and the chances of future occurrences are minimised. Incidents originated by batteries would not normally fall within the reporting criteria of databases such as MHIDAS or MARS, which store information on Major Hazard incidents. In the UK, incidents involving batteries may have been reported under RIDDOR (Reporting of Injuries, Diseases and Dangerous Occurrences Regulations).

4.1. Fire fighting recommendations

The use of water to tackle primary lithium and lithium-ion battery fires was recommended by Farrington (2001) when fires involve small inventories. As the amount of lithium involved would be relatively small in those cases, flammable products and subsequent temperature rise upon ignition would be compensated by the cooling effect of copious amounts of water. Based on experimental evidence from tests carried out by the US Federal Aviation Administration, Reif et al. (2010) recommended that water should not be used if the fire involves primary lithium batteries, to avoid rapid evolution of highly flammable H₂ by reaction of Li metal with water. Halon-based fire extinguishers were also found incapable to extinguish fires involving primary lithium batteries. Fire extinguishing media suggested by Reif et al. (2010) are shown in Table 2.

Table 2 – Firefighting recommendations for dealing with primary lithium and lithium-ion fires (Reif et al., 2010).

	Primary lithium	Lithium-ion
Fires involving batteries only	Lith-X Class D extinguishing agent, no water	ABC dry chemical extinguisher or water
Fires involving batteries and other materials	ABC dry chemical extinguisher or water	ABC dry chemical extinguisher or water, according to combustible materials involved

There is, however, limited information available in the open literature on smoke composition and toxicological properties of smoke generated in primary lithium and lithium-ion battery fires, as well as large testing of fire start and propagation mechanisms under conditions relevant to waste battery storage and recycling operation/processes.

5. Conclusions

A dramatic growth in the use of primary lithium and lithium-ion batteries has taken place since the 1990s as they became the power technologies of choice for portable electronic devices. Growth in the lithium-ion battery market is expected to continue as a new generation of lithium-ion batteries are considered to have the potential to support fully electric vehicles (Cairns and Albertus, 2010).

Primary lithium and lithium-ion batteries may contain a number of hazardous components including lithium metal, and flammable and highly flammable solutions. Faults and misuse during production and operation at their intended application can cause internal and external short-circuits, overcharge, overdischarge or overheating which, if unmanaged, can result in thermal runaway, over-pressurisation, venting, fire and/or explosion of battery packs. Safety tests and control mechanisms have been designed to provide protection against undesired events. The performance of primary lithium and lithium-ion batteries in safety tests markedly differs according to the cell chemistry. New and old batteries have been reported to show significant differences in safety performance under tests conditions. Abuse tests on used batteries may be therefore more suitable to characterise hazards associated with used and waste batteries.

Despite protection provided by a number of safety devices, incidents involving primary lithium and lithium-ion batteries during use, transport, storage and recycling operations continue to take place. There is no common source of information that documents incidents involving these batteries. Such a database would provide an invaluable insight into earlier incident occurrences, facilitating statistical and causal analysis of the events and thus preventing recurrence.

The potential severity of incidents during storage, transport and recycling of waste primary lithium and lithium-ion batteries can be significantly higher than in end-use applications, as significant numbers of batteries may be present at one given location. Treatment and recycling of batteries are usually performed by third-party operators that may have limited access to information on the batteries' history, physical integrity and chemistry. Safe storage, packaging and labelling practices as well as communication among parties involved are essential to ensure safety across the whole lifecycle of primary lithium and lithium-ion batteries.

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